INTELLIGENT VACCINE COLD STORAGE MONITORING SYSTEM INDUSTRY CONTEXT

SCENARIO-BASED REQUIREMENT ANALYSIS

Submitted in the partial fulfilment for the Course of

CSA1016 – SOFTWARE ENGINEERING

to the award of the degree of

BACHELOR OF TECHNOLOGY IN INFORMATION TECHNOLOGY

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July 2026

1. INTRODUCTION

The healthcare industry relies heavily on vaccines to prevent infectious diseases and protect public health. Since vaccines are highly sensitive biological products, they must be stored under controlled environmental conditions throughout manufacturing, transportation, storage, and administration. Any deviation from the recommended storage temperature can reduce vaccine effectiveness, resulting in financial losses and potential health risks.

Traditional vaccine monitoring methods primarily depend on manual temperature recording or standalone monitoring devices. These approaches are often inefficient because they cannot provide continuous monitoring or predict refrigeration failures before they occur. As healthcare organizations expand their operations across multiple facilities, manual monitoring becomes increasingly difficult, creating a need for a more intelligent and automated solution.

The **Intelligent Vaccine Cold Storage Monitoring System** is designed to address these challenges using Internet of Things (IoT), Artificial Intelligence (AI), Cloud Computing, and Predictive Analytics. The proposed software continuously monitors environmental conditions such as temperature, humidity, power supply, and refrigeration health. It predicts equipment failures, generates real-time alerts, maintains historical records, and automatically prepares compliance reports required by healthcare authorities.

This Software Requirement Specification (SRS) document defines the functional requirements, non-functional requirements, system features, actors, use cases, external interfaces, data requirements, assumptions, constraints, and validation criteria necessary for the successful development of the proposed system. The document serves as a communication bridge between stakeholders, software developers, testers, and project managers while ensuring that all software requirements are clearly defined before implementation begins.

2. PROBLEM DESCRIPTION

Vaccines require uninterrupted cold-chain management to preserve their effectiveness. Most vaccines must be stored between **2°C and 8°C**, while certain specialized vaccines require even lower temperatures. Failure to maintain these conditions can permanently reduce vaccine potency, making the vaccines ineffective for patient use.

Many hospitals, clinics, blood banks, pharmacies, and vaccination centers continue to rely on manual temperature monitoring or isolated electronic data loggers. These methods provide limited functionality because they depend on periodic human inspection and often detect temperature excursions only after vaccine damage has already occurred. Human errors, delayed maintenance, power failures, equipment malfunctions, and transportation delays further increase the possibility of vaccine spoilage.

Another significant challenge involves regulatory compliance. Healthcare organizations are required to maintain accurate records of storage conditions, equipment maintenance, calibration reports, and audit logs. Manual documentation is time-consuming and increases the possibility of missing or inaccurate records during inspections.

The proposed Intelligent Vaccine Cold Storage Monitoring System addresses these issues by introducing an automated software platform capable of continuously monitoring environmental conditions through IoT sensors. Artificial Intelligence analyzes real-time sensor data to identify abnormal patterns and predict equipment failures before they occur. Whenever unsafe conditions are detected, the system immediately notifies healthcare personnel through mobile applications, SMS, and email. Historical records, maintenance logs, and automated compliance reports further improve operational efficiency and simplify regulatory inspections.

3. BUSINESS DOMAIN

The proposed system belongs to the **Healthcare Information Technology (HealthTech)** domain with a primary focus on vaccine cold-chain management. It combines modern software technologies such as IoT, Artificial Intelligence, Cloud Computing, Data Analytics, and Secure Web Applications to improve the management of temperature-sensitive medical products.

The software is intended for organizations involved in vaccine storage, transportation, and distribution. These include government hospitals, private hospitals, vaccination centers, blood banks, pharmaceutical warehouses, research laboratories, healthcare logistics providers, and public health agencies.

The business value of the proposed system lies in its ability to reduce vaccine wastage, improve operational efficiency, increase regulatory compliance, and enhance patient safety. By automating environmental monitoring and predictive maintenance, healthcare organizations can reduce financial losses associated with damaged vaccines while ensuring uninterrupted vaccine availability.

The software also supports centralized monitoring across multiple healthcare facilities, allowing administrators to supervise geographically distributed refrigeration units from a single cloud-based dashboard. This capability significantly improves resource utilization and decision-making.

The major business goals of the proposed system include:

- Improving vaccine storage safety.
- Reducing vaccine wastage.
- Enhancing patient safety.
- Increasing equipment reliability.
- Supporting predictive maintenance.
- Automating compliance reporting.
- Reducing operational costs.
- Providing centralized monitoring.
- Improving healthcare efficiency.
- Supporting scalable healthcare infrastructure.

4. STAKEHOLDER ANALYSIS

The Intelligent Vaccine Cold Storage Monitoring System involves multiple stakeholders who interact with the software directly or indirectly. Each stakeholder has specific responsibilities that influence the functional and non-functional requirements of the system. Understanding stakeholder needs is essential for designing software that satisfies business objectives while maintaining healthcare standards.

Healthcare Administrator

Healthcare administrators are responsible for monitoring all vaccine storage facilities within the organization. They supervise system operations, review compliance reports, allocate resources, approve maintenance activities, and ensure that vaccines remain safely stored.

Responsibilities include:

- Monitoring system performance.
- Managing users.
- Reviewing compliance reports.
- Approving maintenance activities.
- Supervising multiple healthcare facilities.

Doctors

Doctors depend on safe and effective vaccines during immunization programs. Although they do not directly interact with storage equipment, they rely on the software to verify vaccine quality before administration.

Responsibilities include:

- Verifying vaccine availability.
- Confirming vaccine safety.
- Reporting damaged vaccine batches.

Nurses

Nurses regularly access vaccine storage units and respond to system notifications. They ensure vaccines remain properly stored and immediately address environmental alerts.

Responsibilities include:

- Monitoring refrigerator conditions.
- Responding to emergency alerts.
- Updating inventory.
- Reporting storage issues.

Pharmacists

Pharmacists manage vaccine inventory while ensuring that stored vaccines remain within recommended environmental conditions.

Responsibilities include:

- Managing inventory.
- Tracking expiry dates.
- Monitoring storage conditions.
- Recording stock movement.

Cold Chain Managers

Cold chain managers supervise storage operations across multiple healthcare facilities. They coordinate equipment maintenance and monitor environmental conditions.

Responsibilities include:

- Supervising refrigeration systems.
- Monitoring equipment performance.
- Coordinating maintenance.
- Reviewing environmental reports.

Maintenance Engineers

Maintenance engineers inspect refrigeration equipment, repair faults, and ensure uninterrupted operation of vaccine storage systems.

Responsibilities include:

- Equipment maintenance.
- Fault diagnosis.
- Preventive maintenance.
- Sensor calibration.

Regulatory Authorities

Government healthcare agencies verify compliance with vaccine storage regulations and inspect audit documentation generated by the system.

Responsibilities include:

- Conducting inspections.
- Reviewing compliance reports.
- Verifying audit records.
- Monitoring healthcare standards.

System Administrator

The system administrator manages cloud servers, databases, software updates, security configurations, and user accounts.

Responsibilities include:

- User management.
- Server monitoring.
- Backup management.
- Security administration.
- Software maintenance.

Software Development Team

The development team designs, develops, tests, deploys, and maintains the software according to stakeholder requirements.

Responsibilities include:

- Requirement analysis.
- Software development.
- Testing.
- Deployment.
- Maintenance.
- Continuous improvement.

Patients

Patients are the ultimate beneficiaries of the proposed system because it ensures that administered vaccines remain safe, effective, and properly stored throughout the cold chain.

The successful implementation of the Intelligent Vaccine Cold Storage Monitoring System depends on effective collaboration among all these stakeholders. Each stakeholder contributes unique functional requirements that collectively influence the overall software architecture, user interface, security policies, reporting capabilities, and operational workflows.

5. EXISTING CHALLENGES

The healthcare industry faces numerous challenges in maintaining the vaccine cold chain. Although refrigeration equipment and manual monitoring procedures have been used for many years, they are no longer sufficient for managing large-scale vaccination programs. The increasing number of healthcare facilities, strict regulatory requirements, and growing dependence on temperature-sensitive vaccines demand an intelligent software solution capable of providing continuous monitoring and predictive decision-making.

One of the primary challenges is the dependence on manual temperature monitoring. Healthcare personnel are required to inspect refrigerator temperatures at regular intervals and record observations in logbooks or spreadsheets. This process is time-consuming, labor-intensive, and highly dependent on human accuracy. Missed observations, incorrect recordings, or delayed inspections may result in vaccines being exposed to unsafe temperatures without immediate detection.

Equipment failure is another significant challenge. Refrigerators and freezers are mechanical systems that may experience compressor failures, electrical faults, refrigerant leakage, or cooling inefficiencies. Since traditional monitoring systems generally detect failures only after temperature deviations occur, vaccines may already be damaged before corrective actions are initiated.

Power interruptions further increase the complexity of vaccine storage management. Unexpected power failures can rapidly increase storage temperatures, particularly in regions where backup power systems are limited. Existing monitoring methods often lack real-time notification capabilities, delaying emergency response.

Healthcare organizations also face challenges in maintaining regulatory compliance. Government agencies require detailed documentation regarding storage conditions, maintenance history, equipment calibration, and audit records. Preparing these reports manually consumes significant time while increasing the possibility of documentation errors.

Another challenge involves the absence of centralized monitoring. Organizations operating multiple hospitals or vaccination centers often maintain independent storage records at each location. Administrators cannot monitor all facilities simultaneously, reducing operational visibility and delaying management decisions.

The growing adoption of digital healthcare systems also introduces cybersecurity concerns. Healthcare data must remain confidential, protected against unauthorized access, and available whenever required. Traditional monitoring solutions often lack advanced security mechanisms needed to protect cloud-based healthcare infrastructure.

The major challenges identified in the existing system include:

- Manual temperature recording.
- Human recording errors.
- Delayed identification of equipment failures.
- Lack of predictive maintenance.
- Poor visibility across multiple facilities.
- Absence of centralized monitoring.
- Inadequate regulatory reporting.
- High vaccine wastage.
- Unexpected refrigeration failures.
- Power outages.
- Sensor failures.
- Delayed emergency response.
- High maintenance costs.
- Limited scalability.
- Weak cybersecurity mechanisms.
- Difficulty analyzing historical data.
- Lack of Artificial Intelligence support.
- Limited automation.
- Poor integration with hospital systems.
- Increased operational workload.

The proposed Intelligent Vaccine Cold Storage Monitoring System addresses these challenges through continuous IoT monitoring, Artificial Intelligence, cloud-based analytics, predictive maintenance, automated notifications, and centralized management.

6. SYSTEM OBJECTIVES

The Intelligent Vaccine Cold Storage Monitoring System has been developed with the objective of ensuring safe, efficient, and intelligent management of vaccine storage across healthcare organizations. The software aims to eliminate the limitations of traditional monitoring methods by introducing real-time data collection, predictive analytics, automated reporting, and cloud-based management.

The primary objective of the proposed system is to maintain uninterrupted vaccine cold-chain conditions by continuously monitoring environmental parameters using Internet of Things (IoT) sensors. Instead of relying on periodic manual inspections, the software automatically collects sensor readings and immediately detects abnormal environmental conditions.

Another important objective is to predict refrigeration failures before they occur. Artificial Intelligence algorithms analyze equipment behavior and historical sensor data to estimate the probability of future failures. This predictive maintenance approach minimizes equipment downtime and reduces vaccine wastage.

The proposed system also aims to simplify healthcare operations by automating maintenance scheduling, compliance reporting, audit documentation, and environmental monitoring. Healthcare personnel receive instant notifications whenever abnormal conditions are detected, allowing immediate corrective actions that protect vaccine quality.

The major objectives of the proposed software are:

- Continuously monitor vaccine storage temperature.
- Monitor humidity levels.
- Monitor refrigerator health.
- Monitor power supply status.
- Monitor battery backup.
- Detect abnormal environmental conditions.
- Predict refrigeration equipment failures.
- Notify healthcare personnel immediately.
- Reduce vaccine wastage.
- Improve patient safety.
- Maintain historical environmental records.
- Generate automated compliance reports.
- Improve maintenance scheduling.
- Reduce operational costs.
- Increase equipment reliability.
- Provide centralized cloud monitoring.
- Support multiple healthcare facilities.
- Improve decision-making using analytics.
- Protect healthcare information.
- Ensure regulatory compliance.

By achieving these objectives, the proposed system significantly enhances vaccine cold-chain management while supporting the long-term digital transformation of healthcare organizations.

7. SCOPE OF THE SYSTEM

The scope of the Intelligent Vaccine Cold Storage Monitoring System covers the continuous monitoring and management of vaccine storage conditions throughout healthcare organizations. The software provides an integrated platform capable of collecting environmental data, analyzing equipment performance, predicting failures, generating alerts, maintaining audit records, and producing compliance reports.

The system is designed for deployment across hospitals, clinics, pharmacies, blood banks, pharmaceutical warehouses, government vaccination centers, research laboratories, and vaccine distribution facilities. Since the software operates on cloud infrastructure, administrators can monitor geographically distributed storage units through a centralized dashboard without visiting individual locations.

The proposed software continuously collects data from IoT sensors installed inside refrigeration units. These sensors monitor environmental parameters including temperature, humidity, refrigerator door status, compressor health, power supply condition, battery backup status, and overall equipment performance. The collected information is securely transmitted to cloud servers for storage, analysis, and visualization.

Artificial Intelligence continuously analyzes historical and real-time sensor data to identify abnormal operating patterns, estimate equipment health, and predict future refrigeration failures. The notification engine immediately alerts healthcare personnel whenever environmental conditions become unsafe.

Apart from monitoring, the system also manages equipment maintenance schedules, historical logs, user activities, compliance documentation, inventory information, and performance reports. Authorized users can generate customized reports for regulatory inspections, organizational analysis, and management decision-making.

The scope of the proposed software includes:

- Real-time environmental monitoring.
- Temperature monitoring.
- Humidity monitoring.
- Power supply monitoring.
- Refrigerator health monitoring.
- IoT sensor management.
- Artificial Intelligence prediction.
- Predictive maintenance.
- Emergency alert generation.
- Mobile application support.
- Web dashboard.
- Historical data storage.
- Equipment maintenance management.

- Vaccine inventory monitoring.
- Compliance report generation.
- Audit trail management.
- User authentication.
- Role-based access control.
- Cloud database management.
- Data analytics and visualization.

The proposed system does not perform vaccine manufacturing, transportation management, or patient immunization activities. Its primary responsibility is to ensure that vaccines remain under recommended storage conditions throughout the cold-chain process while supporting healthcare personnel with intelligent monitoring and decision-making capabilities.

8. FUNCTIONAL REQUIREMENTS

Functional requirements describe the services and operations that the Intelligent Vaccine Cold Storage Monitoring System must perform. These requirements define the expected behavior of the software from the user's perspective and ensure that stakeholder expectations are satisfied.

The proposed system shall continuously collect environmental information from IoT sensors installed inside vaccine storage units. Temperature, humidity, power supply status, compressor health, battery backup, and refrigerator door status shall be monitored continuously and stored securely in the cloud database.

Artificial Intelligence shall analyze environmental data to identify abnormal patterns and predict equipment failures before they occur. Whenever unsafe conditions are detected, the notification module shall automatically inform authorized healthcare personnel through multiple communication channels.

The software shall provide a centralized dashboard that displays real-time sensor readings, equipment status, maintenance schedules, environmental trends, and active alerts. Healthcare administrators shall be able to monitor multiple healthcare facilities simultaneously using this dashboard.

The reporting module shall automatically generate audit logs, maintenance records, compliance reports, historical temperature reports, and equipment performance summaries. The software shall also maintain historical environmental data for future analysis.

The major functional requirements of the proposed system are listed below.

User Management

- User registration.
- Secure login.
- Password reset.
- Multi-factor authentication.
- Role-based access control.
- User profile management.

Sensor Management

- Register IoT sensors.
- Read temperature.
- Read humidity.
- Monitor power supply.
- Monitor compressor health.
- Monitor battery backup.
- Detect sensor failure.

Monitoring

- Display live temperature.
- Display humidity.
- Display equipment health.
- Display environmental graphs.
- Monitor multiple storage units.
- Monitor multiple hospitals.

Artificial Intelligence

- Detect anomalies.
- Predict equipment failure.
- Generate equipment health score.
- Analyze historical trends.
- Estimate maintenance requirements.

Notification

- SMS alerts.
- Email alerts.
- Mobile push notifications.
- Emergency notifications.
- Escalation alerts.

Reporting

- Compliance reports.
- Maintenance reports.
- Audit reports.
- Historical temperature reports.
- Equipment performance reports.
- PDF export.
- Excel export.

Maintenance

- Schedule maintenance.
- Assign technicians.
- Record repairs.
- Store maintenance history.
- Generate maintenance reminders.

Inventory

- Monitor vaccine inventory.
- Track expiry dates.
- Record stock movement.
- Generate inventory reports.

Cloud Administration

- Cloud database management.
- Data backup.
- Disaster recovery.
- API management.
- Performance monitoring.
- Security management.

The functional requirements described above establish the core capabilities of the proposed Intelligent Vaccine Cold Storage Monitoring System and provide a solid foundation for subsequent software design, implementation, testing, and deployment.

9. NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements define the quality attributes of the proposed system. They ensure the software performs efficiently, securely, and reliably under different operating conditions.

Performance

- Dashboard response time should be less than 2 seconds.
- Sensor data should update in real time.
- Support multiple users simultaneously.

Security

- User authentication and authorization.
- Role-Based Access Control (RBAC).
- Data encryption using HTTPS/TLS.
- Secure cloud storage.

Reliability

- Continuous system monitoring.
- Automatic data backup.
- Fault tolerance during failures.

Scalability

- Support thousands of IoT sensors.
- Allow multiple hospitals to use the system.
- Easy addition of new storage units.

Usability

- Simple user interface.
- Responsive web and mobile applications.
- Easy navigation and report generation.

Maintainability

- Modular architecture.
- Reusable code.
- Easy software updates and debugging.

Availability

- 99.9% system uptime.
- Cloud-based deployment.
- Disaster recovery support.

10. ACTOR IDENTIFICATION

The system contains both primary and secondary actors.

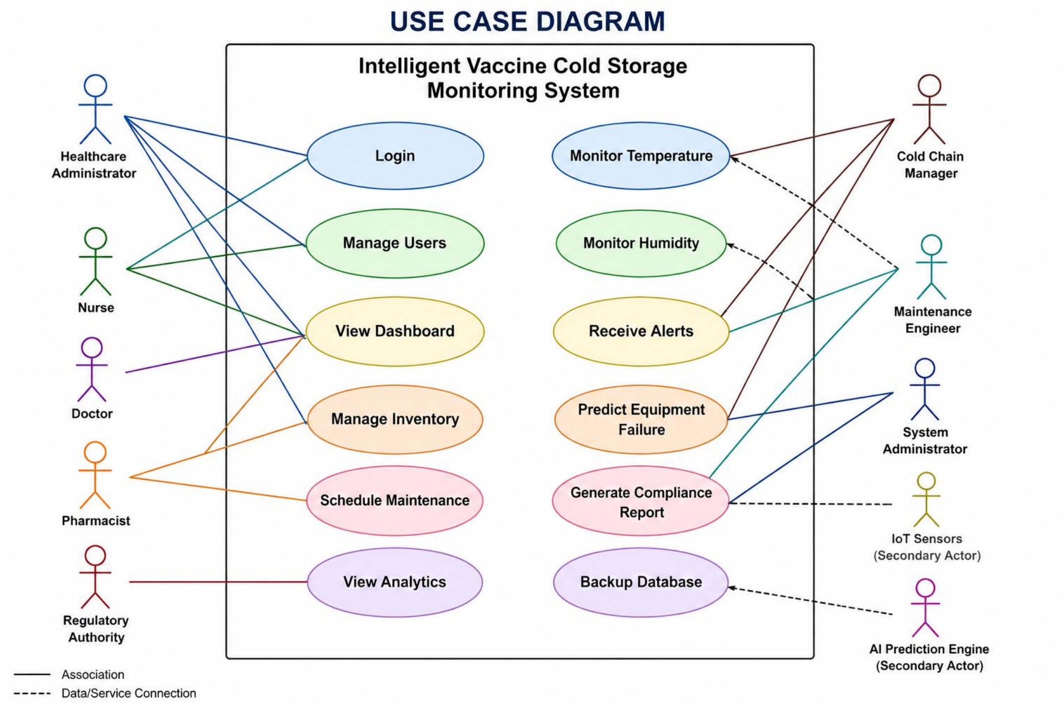
Actor	Type	Major Use Cases
Healthcare Administrator	Primary	Login, Monitor Dashboard, Generate Reports
Doctor	Primary	View Vaccine Status
Nurse	Primary	Monitor Temperature, Receive Alerts
Pharmacist	Primary	Manage Inventory
Cold Chain Manager	Primary	Monitor Equipment
Maintenance Engineer	Primary	View Maintenance Schedule
Regulatory Authority	Secondary	View Compliance Reports
System Administrator	Primary	Manage Users, Backup Database
AI Prediction Engine	Secondary	Predict Equipment Failure
IoT Sensors	Secondary	Send Temperature and Humidity Data

11. USE CASES

The major use cases of the system are:

- User Login
- Manage Users
- Monitor Temperature
- Monitor Humidity
- Display Dashboard
- Receive Alerts
- Predict Equipment Failure
- Schedule Maintenance
- Generate Reports
- Manage Vaccine Inventory
- Backup Database
- View Analytics

Use Case Diagram



12. USE CASE DESCRIPTIONS

Use Case 1 – User Login

Actor: Administrator

Description: Authenticates authorized users.

Pre-condition: User account exists.

Post-condition: Dashboard is displayed.

Use Case 2 – Monitor Temperature

Actor: Nurse

Description: Displays live temperature of storage units.

Pre-condition: Sensor connected.

Post-condition: Latest temperature shown.

Use Case 3 – Receive Alert

Actor: Nurse

Description: Sends emergency notification when unsafe temperature is detected.

Pre-condition: Temperature exceeds threshold.

Post-condition: Notification delivered.

Use Case 4 – Generate Report

Actor: Administrator

Description: Generates compliance reports.

Pre-condition: Data available.

Post-condition: PDF report generated.

Use Case 5 – Predict Equipment Failure

Actor: AI Engine

Description: Predicts refrigeration failure.

Pre-condition: Historical data exists.

Post-condition: Risk score generated.

13. DATA REQUIREMENTS

The system stores operational and historical information for monitoring and reporting purposes.

User Data

- User ID
- Name
- Email
- Password
- Role

Sensor Data

- Sensor ID
- Temperature
- Humidity
- Timestamp
- Status

Equipment Data

- Refrigerator ID
- Health Status
- Power Status
- Maintenance Date

Inventory Data

- Vaccine Name
- Batch Number
- Quantity
- Expiry Date

Alert Data

- Alert ID
- Time
- Severity
- Status

14. EXTERNAL INTERFACE REQUIREMENTS

User Interface

- Web Dashboard
- Android Application
- iOS Application

Hardware Interface

- Temperature Sensors
- Humidity Sensors
- Power Sensors
- Refrigerator Controllers

Software Interface

- Cloud Database
- REST APIs
- AI Prediction Module
- Notification Service

Communication Interface

- HTTPS
- Wi-Fi
- MQTT
- SMS Gateway
- Email Server

15. SYSTEM FEATURES

The Intelligent Vaccine Cold Storage Monitoring System provides the following features:

- Continuous temperature monitoring
- Humidity monitoring
- Refrigerator health monitoring
- AI-based failure prediction
- Automated notifications
- Cloud dashboard
- Historical data analysis
- Compliance report generation

- Maintenance scheduling
- Vaccine inventory management
- User authentication
- Secure cloud storage
- Audit logs
- Data backup
- Multi-hospital support

16. ASSUMPTIONS

The following assumptions are considered during system development:

- Internet connectivity is available.
- IoT sensors function correctly.
- Cloud servers remain operational.
- Healthcare staff are trained to use the system.
- Refrigeration equipment supports sensor installation.
- Users possess valid login credentials.

17. CONSTRAINTS

The proposed system is affected by several constraints.

Technical Constraints

- IoT sensor compatibility.
- Internet dependency.
- Cloud infrastructure.

Operational Constraints

- Staff training required.
- Periodic sensor calibration.

Economic Constraints

- Initial deployment cost.
- Maintenance cost.

Legal Constraints

- Healthcare regulations.
- Data privacy laws.

Time Constraints

- Agile sprint deadlines.
- Limited implementation schedule.

18. REQUIREMENT VALIDATION

The requirements were validated to ensure completeness, consistency, and feasibility.

Completeness

- Functional requirements identified.
- Non-functional requirements identified.
- Stakeholders covered.
- Use cases defined.

Consistency

- No conflicting requirements.
- Modules support overall objectives.
- Stakeholder needs are aligned.

Ambiguity Check

- Requirements use clear language.
- Technical terms are defined.
- User roles are specified.

Suggested Improvements

- Introduce Blockchain for secure audit logs.
- Add Edge AI for faster predictions.
- Integrate with Hospital Information Systems.
- Improve multilingual support.
- Develop offline monitoring capability.

19. CONCLUSION

The Software Requirement Specification (SRS) for the Intelligent Vaccine Cold Storage Monitoring System defines the functional and non-functional requirements necessary for developing a secure, reliable, and scalable healthcare application. The proposed solution integrates IoT, Artificial Intelligence, and cloud technologies to improve vaccine storage management, reduce wastage, and ensure regulatory compliance. By clearly identifying stakeholders, system features, use cases, assumptions, and constraints, the SRS provides a strong foundation for software design, implementation, testing, and deployment.

20. REFERENCES (IEEE Format)

[1] I. Sommerville, *Software Engineering*, 10th ed. Pearson, 2016.

[2] R. S. Pressman and B. R. Maxim, *Software Engineering: A Practitioner's Approach*, 9th ed. McGraw-Hill, 2020.

[3] World Health Organization, *Temperature Sensitivity of Vaccines*, Geneva, Switzerland, 2006.

[4] Centers for Disease Control and Prevention, *Vaccine Storage and Handling Toolkit*, Atlanta, USA, 2023.

[5] ISO 13485:2016, *Medical Devices—Quality Management Systems*, ISO, Geneva, Switzerland, 2016.

[6] ISO/IEC 27001:2022, *Information Security Management Systems*, ISO, Geneva, Switzerland, 2022.

[7] A. Bahga and V. Madisetti, *Internet of Things: A Hands-On Approach*. Universities Press, 2015.

[8] M. Kleppmann, *Designing Data-Intensive Applications*. O'Reilly Media, 2017.

[9] J. Dean and A. Rajkomar, "Machine Learning in Medicine," *New England Journal of Medicine*, vol. 380, no. 14, pp. 1347–1358, 2019.

[10] IEEE Computer Society, *IEEE Software Engineering Standards Collection*, IEEE, 2019.